Possible Extensions of Multiple Linear Regression(MLR)

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1 Introduction

Presently, researchers are giving attention on Longitudinal study(panel study) that involve repeated observations of the same variables (e.g., people) over long periods of time. Hence,the data that is collected repeatedly over time, are called longitudinal data(panel data) when repeated measurements are not necessarily collected over time such type of data is called **repeated measures data** So Longitudinal data, Multivariate data, and repeated measurement data are all examples of **correlated data**.
An example of the longitudinal data, Measurements of individuals blood pressures over time. An important feature of longitudinal data is that the repeated measurements of a variable on each individual are likely to be **correlated's** that sure the **efficient** or unbiased results in data analysis. A main advantage of longitudinal studies is that they allow us to study changes over time for variables of interests.

1.1 Models for Longitudinal Data

Regression models are widely used to understand the systematic variation between individuals by the researchers in longitudinal studies in which, the response data are longitudinal measurements on a variable of interest while covariates or predictors can either be longitudinal data (time-dependent covariates, including time) or cross-sectional data (time-independent covariates).

- **Difference between a longitudinal study and a cross-sectional study**

longitudinal study	**cross-sectional study**
Repeated observations	Observations at a single point in time
Observes the same group multiple times	Observes different groups (a cross-section) in the population
Follows changes in participants over time	Provides snapshot of society at a given point

assumes that the repeated measurements within an individual are correlated because these measurements share the **same unobserved characteristics** of the individual. The unobserved characteristics of an individual can be represented by **random effects** , or individual effects, of that individual. Thus, regression models for longitudinal data can be obtained by introducing random effects in the corresponding models for cross-sectional data, leading to random effects models or **mixed effects models**.

1.2 Extension of simple Linear regression model to Linear Mixed Effects(LME) Models

Let y_{ij} be the response value for individual i at time t_{ij}, $i = 1, 2, ..., n, j = 1, 2, ..., n_i$ and **simple linear regression** model for longitudinal data as below

$$y_{ij} = \beta_0 + \beta_1 t_{ij} + e_{ij}, \quad i = 1, 2, ..., n, j = 1, 2, ..., n_i$$

where e_{ij} is a random error. If the data show that β_0 the intercept varies substantially for different individuals, we may introduce a **random effect** for the interceptβ_0 and get **LME model** as below

$$y_{ij} = (\beta_0 + r_{0i}) + \beta_1 t_{ij} + e_{ij}, \quad i = 1, 2, ..., n, j = 1, 2, ..., n_i \tag{1}$$

where r_{0i} is a **random effect** for individual i, $\beta_{0i} = \beta_0 + r_{0i}$ is the individual-specific intercept for individual i, and the slope β_1 is assumed to be the same (fixed) for all individuals. The parameters β_0 and β_1 are called **fixed effects** or population parameters, since they are fixed for all individuals. We often assume that e_{ij} i.i.d $\sim N(0, \sigma^2)$ and $r_{0i} \sim N(0, r^2)$, and e_{ij} and r_{0i} are independent.

Note 1: We can choose random effects formally and informally way from the data set. Firstly, its choose based on standard model selection methods such as AIC or BIC criteria or on standard tests such as the likelihood ratio test. Secondly, the heterogeneous feature of the data use to choose random effect as a informal way.

Matrix form (**General**)of LME models can be written as below:

Let $y_i = (y_{i1}, y_{i2}, ..., y_{in_i})^\top$ be the n_i repeated measurements of the response variable y on individual i, and let $e_i = (e_{i1}, e_{i2}, ..., e_{in_i})^\top$ be the corresponding random errors of the repeated measurements, $i = 1, 2, ..., n$. The general form of LME models is

$$\begin{aligned} y_i &= \boldsymbol{X_i\beta} + W_i\boldsymbol{r_i} + e_i, \quad i = 1, 2, ..., n \qquad (2) \\ \boldsymbol{r_i} &\sim N(0, D^*), \quad \boldsymbol{e_i} \sim N(0, R_i) \qquad (3) \end{aligned}$$

where $\boldsymbol{\beta} = (\beta_1, \beta_2, ..., \beta_p)^\top$ is a $p \times 1$ vector of fixed effects, $\boldsymbol{r_i} = (r_{i1}, ...r_{iq})^\top$ is a $q \times 1$ vector of random effects, matrix $\boldsymbol{X_i}(n_i \times p)$ and matrix $W_i(n_i \times q)$ are known design matrices which often contain covariates (including times), D^* is a $q \times q$ covariance matrix of the random effects, and R_i is a $n_i \times n_i$ covariance matrix of the within-individual random errors.

Note 2: The standard assumptions for models (2) and (3) are

- The individuals are independent.
- The random effects $\boldsymbol{r_i}$ and the errors e_i have mean zero.
- The random effects $\boldsymbol{r_i}$ and the errors e_i are independent and both are normally distributed.

Note 3:

- In LME model (2) and (3), the repeated measurements $\{y_{i1}, y_{i2}, ..., y_{in_i}\}$ of the response can be taken at different time points for different individuals, and the number of measurements n_i may vary across individuals. Thus, a LME model allows unbalanced(missing) data in the response.
- LME models differ from linear regression models for cross-sectional data by the term $W_i\boldsymbol{r_i}$.
- In LME model (2) and (3), we assume $R_i = \sigma^2 I_{n_i}$ that helps to reduces the number of parameters in the model.
- Statistical inference for LME models is typically based on the **maximum likelihood method**. MLE of the fixed parameters $\boldsymbol{\beta}$ can then be obtained using an iterative algorithm such as an expectation-maximization (EM) algorithm or a **Newton-Raphson method**.

Note 4:

- **Examples of Rs formula notation for fixed effects.**

Formula	**Description**
$a + b$	main effects of a and b (and no interaction)
$a : b$	only interaction of a and b (and no main effects)
$a * b$	main effects and interaction of a and b (expands to: $a + b + a : b$)
$(a + b + c)^2$	main effects and two-way interactions, but no three-way interaction (expands to: $a + b + c + a : b + b : c + a : c$)
$(a + b) * c$	all main effects and pairwise interactions between c and a or b (expands to:$a + b + c + a : c + b : c$)
$0 + a$	0 suppresses the intercept resulting in a model that has one parameter per level of a (identical to: $a - 1$)

Note 5:

- **Readers required to see Examples of lme4s formula notation for random effects that described as below**

Formula	Description
$(1\|s)$	random intercepts for unique level of the factor s
$(1\|s) + (1\|i)$	random intercepts for each unique level of s and for each unique level of i
$(1\|s/i)$	random intercepts for factor s and i, where the random effects for i are nested in s.
$(a\|s)$	random intercepts and random slopes for a, for each level of s. Correlations between the intercept and slope effects are also estimated
$(a * b\|s)c$	random intercepts and slopes for a, b, and the $a : b$ interaction, for each level of s. Correlations between all the random effects are estimated
$(0 + a\|s)$	random slopes for a for each level of s, but no random intercepts
$(a\|\|s)$	random intercepts and random slopes for a , for each level of s, but no correlations between the random effects

Example 1: To study the growth of children, the distance from the pituitary gland to the pterygomaxillary fissure is measured every two years from 8 years of age until 14 years of age. A sample of 27 children 16 males and 11 females was obtained by orthodontists from x-rays of the childrens skulls.(**This data set is available in r-package "nlme" by the name of "Orthodont"**) **Solution:**

1. First get the "nlme" and "lattice" library for mixed effects models and see the data set structure as below

```
 > rm(list=ls())
> library(nlme)
> library(lattice)
> data("Orthodont")
> head(Orthodont)
Grouped Data: distance ~ age | Subject
  distance age Subject  Sex
1     26.0   8     M01 Male
2     25.0  10     M01 Male
3     29.0  12     M01 Male
4     31.0  14     M01 Male
5     21.5   8     M02 Male
6     22.5  10     M02 Male
```

2. fit a LME model with random effects on both the intercept and the slope.

```
 > mode1 <- lme(distance~age, data=Orthodont, random = ~age | Subject, method = "ML")
> summary(mode1)
Linear mixed-effects model fit by maximum likelihood
 Data: Orthodont
       AIC      BIC    logLik
  451.2116 467.3044 -219.6058

Random effects:
 Formula: ~age | Subject
 Structure: General positive-definite, Log-Cholesky parametrization
            StdDev    Corr
(Intercept) 2.1940996 (Intr)
age         0.2149244 -0.581
Residual    1.3100399

Fixed effects: distance ~ age
                Value Std.Error DF   t-value p-value
(Intercept) 16.761111 0.7678975 80 21.827278       0
age          0.660185 0.0705779 80  9.353997       0
```

3. **Add Sex and interaction**

```
  > mode2 <- update(mode1, fixed=distance~Sex*age)
> summary(mode2)
Linear mixed-effects model fit by maximum likelihood
 Data: Orthodont
      AIC     BIC   logLik
  443.806 465.263 -213.903

Random effects:
 Formula: ~age | Subject
 Structure: General positive-definite, Log-Cholesky parametrization
            StdDev   Corr
(Intercept) 2.134688 (Intr)
age         0.154139 -0.603
Residual    1.310040

Fixed effects: distance ~ Sex + age + Sex:age
                   Value Std.Error DF   t-value p-value
(Intercept)   16.340625 0.9987521 79 16.361042  0.0000
SexFemale      1.032102 1.5647438 25  0.659598  0.5155
age            0.784375 0.0843294 79  9.301322  0.0000
SexFemale:age -0.304830 0.1321188 79 -2.307238  0.0237
```

Results showed the significant difference between boys and girls

4. **ANOVA test for comparing the two models**

```
  > anova(mode1, mode2)
       Model df      AIC      BIC    logLik   Test  L.Ratio p-value
mode1     1  6 451.2116 467.3044 -219.6058
mode2     2  8 443.8060 465.2630 -213.9030 1 vs 2 11.40565  0.0033
```

mode2 fits much better, so we should add Sex and interaction in the model

5. **Model diagnostics for the LME fit for mode2**

```
###### Check residual plot####
plot(mode2, resid(.,type="p")~fitted(.) | Sex,id=0.05,adj=-0.3,col = "red" )
 # Check normality assumption for errors
qqnorm(mode2, ~resid(.) | Sex, col = "red")
##### Check normality assumption random effects #####
qqnorm(mode2, ~ranef(.),id=0.1,cex=0.7,col = "red" )
```

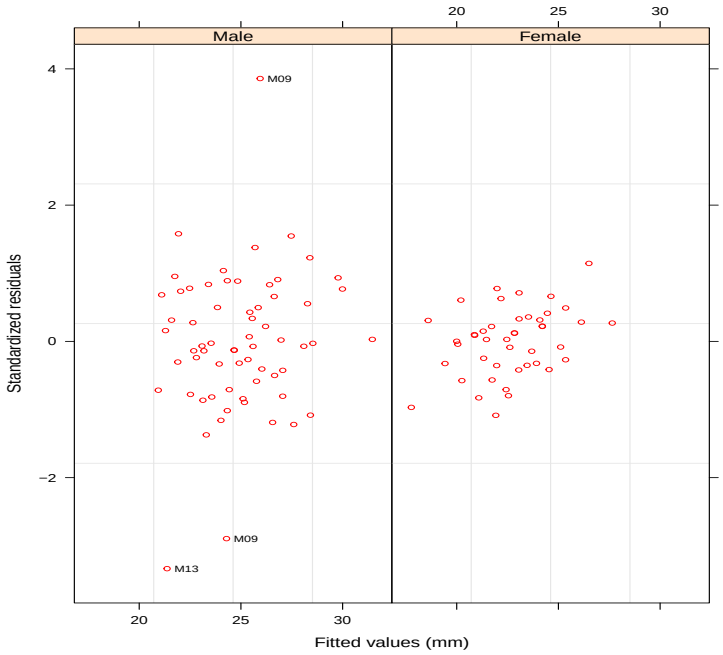


Figure 1: Residual plots

Normality assumptions are good for random effects and within subject errors but residual plot may have an outliers and not effect Model diagnostics for the LME fit.

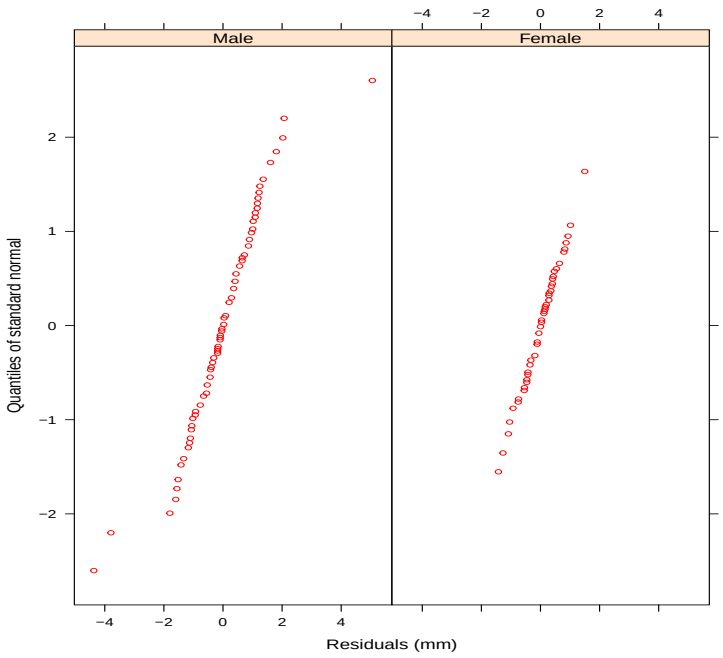


Figure 2: QQ plot for within-individual random errors

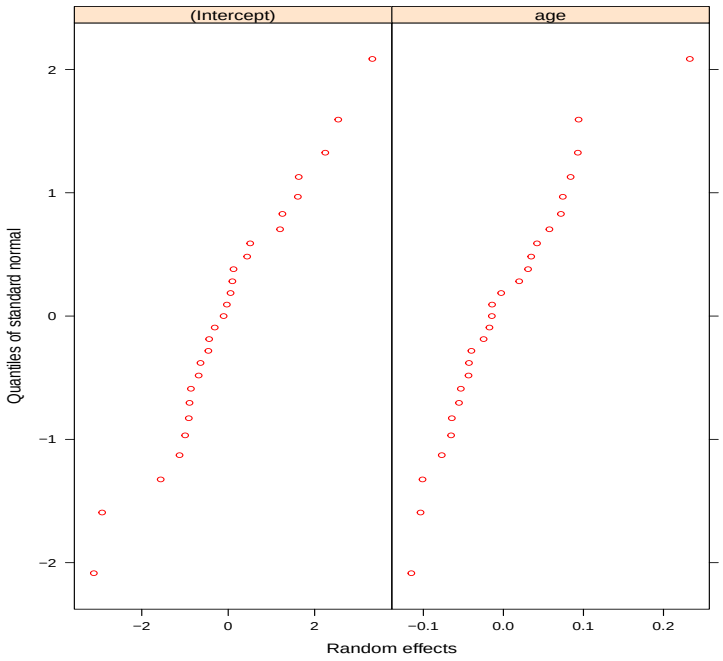


Figure 3: QQ plot for the random effects